14.6 Solution: (a) Similar to Problem 14.5, we can express the energy conservation condition as

$$E = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + V(r) = \text{const.}$$

in polar coordinates. Due to the conservation of angular momentum, we know $L = mr^2\dot{\theta}$ is also a constant. Then, the energy conservation condition becomes

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} + V(r),$$

or

$$\frac{dr}{dt} = \sqrt{\frac{2}{m}}\left[E - V(r) - \frac{L^2}{2mr^2}\right]^{1/2}.$$

Then, as in Problem 14.5 (a), the total energy loss becomes

$$\Delta W = \frac{4}{3}\frac{z^2e^2}{m^2c^3}\sqrt{\frac{m}{2}}\int_{r_{\text{min}}}^{+\infty}\left|\frac{\partial V}{\partial r}\right|^2\left[E - V(r) - \frac{L^2}{2mr^2}\right]^{-1/2} dr,$$

where r_{min} is the solution to

$$E = \frac{L^2}{2mr^2} + V(r).$$

(b) For Coulomb potential,

$$V(r) = \frac{zZe^2}{r},$$

the total energy loss is

$$\Delta W = \frac{4}{3}\frac{z^4Z^2e^6}{m^2c^3}\sqrt{\frac{m}{2}}\int_{r_{\text{min}}}^{+\infty}\frac{1}{r^4}\left[\frac{1}{2}mv_0^2 - \frac{zZe^2}{r} - \frac{mb^2v_0^2}{2r^2}\right]^{-1/2} dr,$$

where we have used the constants of motion $E = mv_0^2/2$ and $L = mbv_0$. This can be further transformed into the following form,

$$\Delta W = \frac{4}{3}\frac{z^4Z^2e^6}{m^2c^3v_0}\int_{r_{\text{min}}}^{+\infty}\frac{1}{r^4}\left[1 - \frac{2zZe^2}{mv_0^2r} - \frac{b^2}{r^2}\right]^{-1/2} dr.$$

Let

$$t = \frac{bmv_0^2}{zZe^2},$$

we then have

$$\Delta W = \frac{4}{3}\frac{zmv_0^5}{Zc^3t^3}\cdot b^3\int_{r_{\text{min}}}^{+\infty}\frac{1}{r^4}\left[1 - \frac{2b}{tr} - \frac{b^2}{r^2}\right]^{-1/2} dr.$$

It is now more convenient to perform the integral in r^{-1}. To this end, we can rewrite the integral as

$$\begin{aligned}
\Delta W &= -\frac{4}{3}\frac{zmv_0^5}{Zc^3t^3}\cdot b^3\int_{r_{\text{min}}}^{+\infty}\frac{1}{r^2}\left[1 - \frac{2b}{tr} - \frac{b^2}{r^2}\right]^{-1/2} d\left(\frac{1}{r}\right) \\
&= \frac{4}{3}\frac{zmv_0^5}{Zc^3t^3}\cdot b^3\int_0^{r_{\text{min}}^{-1}}\frac{x^2}{\sqrt{1 - 2bx/r - b^2x^2}}dx
\end{aligned}$$

$$= \frac{4}{3}\frac{zmv_0^5}{Zc^3t^3}\int_0^{br_{\min}^{-1}} \frac{x^2}{\sqrt{1-2x/t-x^2}}dx.$$

Notice that the quadratic equation

$$1 - \frac{2x}{t} - x^2 = 0$$

should have only one positive solution, which corresponds to

$$x_{\max} = \frac{b}{r_{\min}} = -\frac{1}{t} + \sqrt{1+\frac{1}{t^2}}.$$

Using the integral identities involving quadratic forms (for example, see here),

$$\int \frac{x^2dx}{\sqrt{ax^2+bx+c}} = \frac{2ax-3b}{4a^2}\sqrt{ax^2+bx+c} + \frac{3b^2-4ac}{8a^2}\int \frac{dx}{\sqrt{ax^2+bx+c}} + C,$$

and

$$\int \frac{dx}{\sqrt{ax^2+bx+c}} = \frac{1}{\sqrt{-a}} \arcsin\left(\frac{2ax+b}{\sqrt{b^2-4ac}}\right) + C,$$

for $a < 0$ and the discriminant $b^2 - 4ac > 0$. In our case, $a = -1$, $b = -2/t$, and $c = 1$, we finally have

$$\int \frac{x^2dx}{\sqrt{1-2x/t-x^2}} = \left(-\frac{x}{2}+\frac{3}{2t}\right)\sqrt{1-\frac{2x}{t}-x^2} + \left(\frac{1}{2}+\frac{3}{2t^2}\right)\arcsin\left(\frac{x+1/t}{\sqrt{1+1/t^2}}\right) + C.$$

Then the definite integral can be evaluated as

$$\begin{aligned}
\int_0^{x_{\max}} \frac{x^2}{\sqrt{1-2x/t-x^2}}dx &= \left[\left(-\frac{x}{2}+\frac{3}{2t}\right)\sqrt{1-\frac{2x}{t}-x^2} + \left(\frac{1}{2}+\frac{3}{2t^2}\right)\arcsin\left(\frac{x+1/t}{\sqrt{1+1/t^2}}\right)\right]\Bigg|_0^{x_{\max}} \\
&= -\frac{3}{2t} + \left(\frac{1}{2}+\frac{3}{2t^2}\right)\left[\frac{\pi}{2} - \arctan\left(\frac{1}{t}\right)\right] \\
&= -\frac{3}{2t} + \left(\frac{1}{2}+\frac{3}{2t^2}\right)\arctan t,
\end{aligned}$$

where in the last step, we have used

$$\arctan t + \arctan\left(\frac{1}{t}\right) = \frac{\pi}{2}.$$

Finally,

$$\begin{aligned}
\Delta W &= \frac{4}{3}\frac{zmv_0^5}{Zc^3t^3}\left[-\frac{3}{2t} + \left(\frac{1}{2}+\frac{3}{2t^2}\right)\arctan t\right] \\
&= \frac{2zmv_0^5}{Zc^3}\left[-\frac{1}{t^4} + \frac{1}{t^5}\left(1+\frac{t^2}{3}\right)\arctan t\right].
\end{aligned}$$

(c) For $t = \cot(\theta/2)$, we have

$$\arctan t = \frac{\pi}{2} - \frac{\theta}{2}.$$

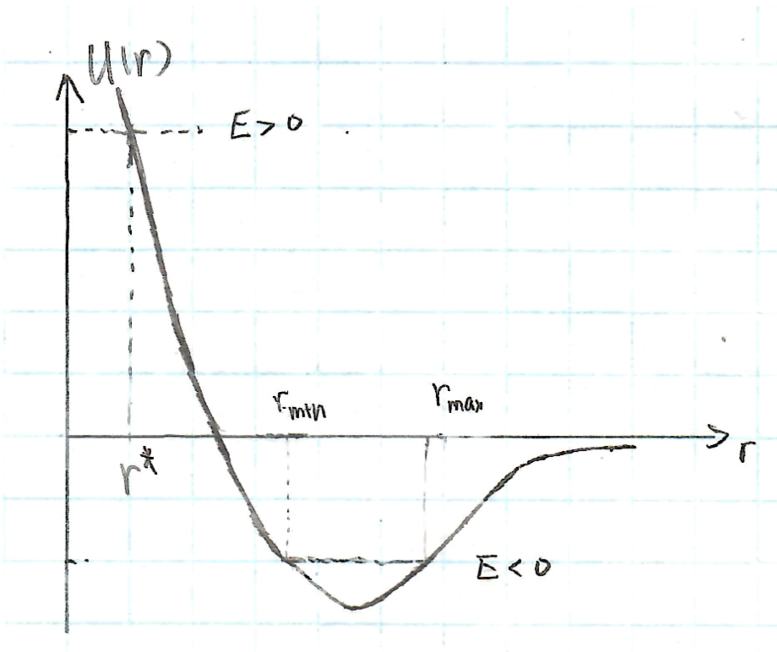

Figure 1: Effective potential for attractive interaction.

Then,

$$\begin{aligned}
\Delta W &= \frac{2zmv_0^5}{Zc^3}\left[-\tan^4\left(\frac{\theta}{2}\right)+\tan^3\left(\frac{\theta}{2}\right)\left(\tan^2\left(\frac{\theta}{2}\right)+\frac{1}{3}\right)\cdot\frac{1}{2}(\pi-\theta)\right] \\
&= \frac{2zmv_0^5}{Zc^3}\tan^3\left(\frac{\theta}{2}\right)\left[\frac{1}{6}(\pi-\theta)\left(1+3\tan^2\left(\frac{\theta}{2}\right)\right)-\tan\left(\frac{\theta}{2}\right)\right].
\end{aligned}$$

(d) The effective potential of an attractive Coulomb interaction is given by (Fig. (1))

$$U(r) = -\frac{zZe^2}{r}+\frac{L^2}{2mr^2},$$

see *Mechanics*, Landau and Lifshitz, Section 15. When the total energy $E = m\dot{r}^2/2 + U(r)$ is positive, the particle's trajectory is unbounded, $r > r^*$. Then, the total radiation energy loss can be similarly calculated. However, if the total energy of the particle is negative, the particle is bounded to the central potential. During its periodic motion between $r_{\min}$ and $r_{\max}$, it will constantly lose energy due to the radiation. Eventually, it will lose all of its energy and fall to the center.